Database operations on EV Charging dataset

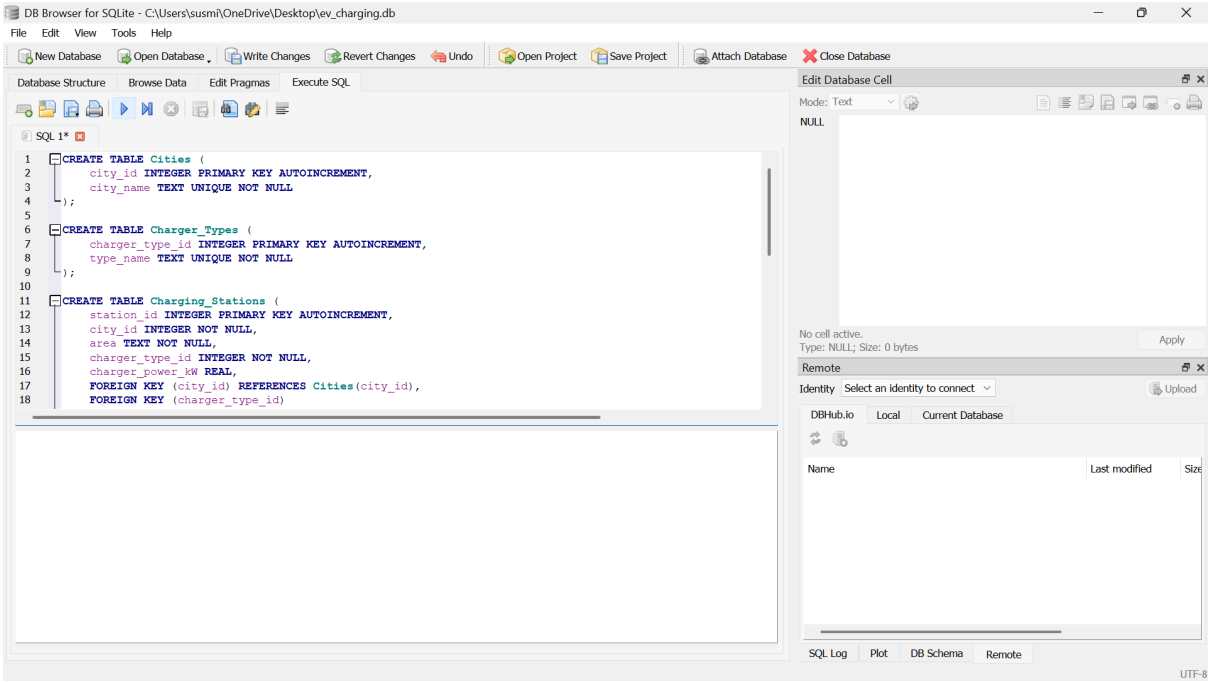
```sql
CREATE TABLE Cities (
    city_id INTEGER PRIMARY KEY AUTOINCREMENT,
    city_name TEXT UNIQUE NOT NULL
);

CREATE TABLE Charger_Types (
    charger_type_id INTEGER PRIMARY KEY AUTOINCREMENT,
    type_name TEXT UNIQUE NOT NULL
);

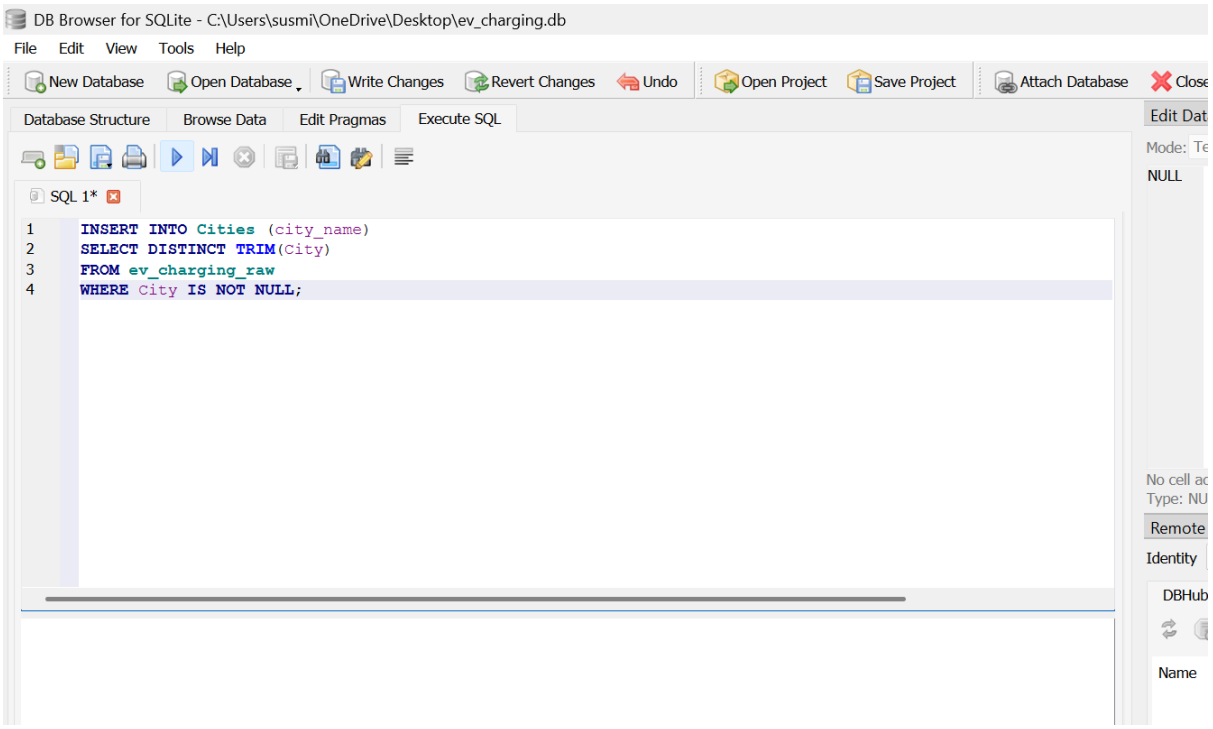
CREATE TABLE Charging_Stations (
    station_id INTEGER PRIMARY KEY AUTOINCREMENT,
    city_id INTEGER NOT NULL,
    area TEXT NOT NULL,
    charger_type_id INTEGER NOT NULL,
    charger_power_kW REAL,
    FOREIGN KEY (city_id) REFERENCES Cities(city_id),
    FOREIGN KEY (charger_type_id)
        REFERENCES Charger_Types(charger_type_id)
);

CREATE TABLE Charging_Sessions (
    session_id TEXT PRIMARY KEY,
    station_id INTEGER NOT NULL,
    session_start TEXT,
    energy_kWh REAL,
    duration_minutes REAL,
    wait_minutes INTEGER,
    battery_end_percent INTEGER,
    co2_saved_kg REAL,
    customer_rating REAL,
    grid_load_impact_kW REAL,
    charging_efficiency REAL,
    session_status TEXT,
    FOREIGN KEY (station_id)
        REFERENCES Charging_Stations(station_id)
);

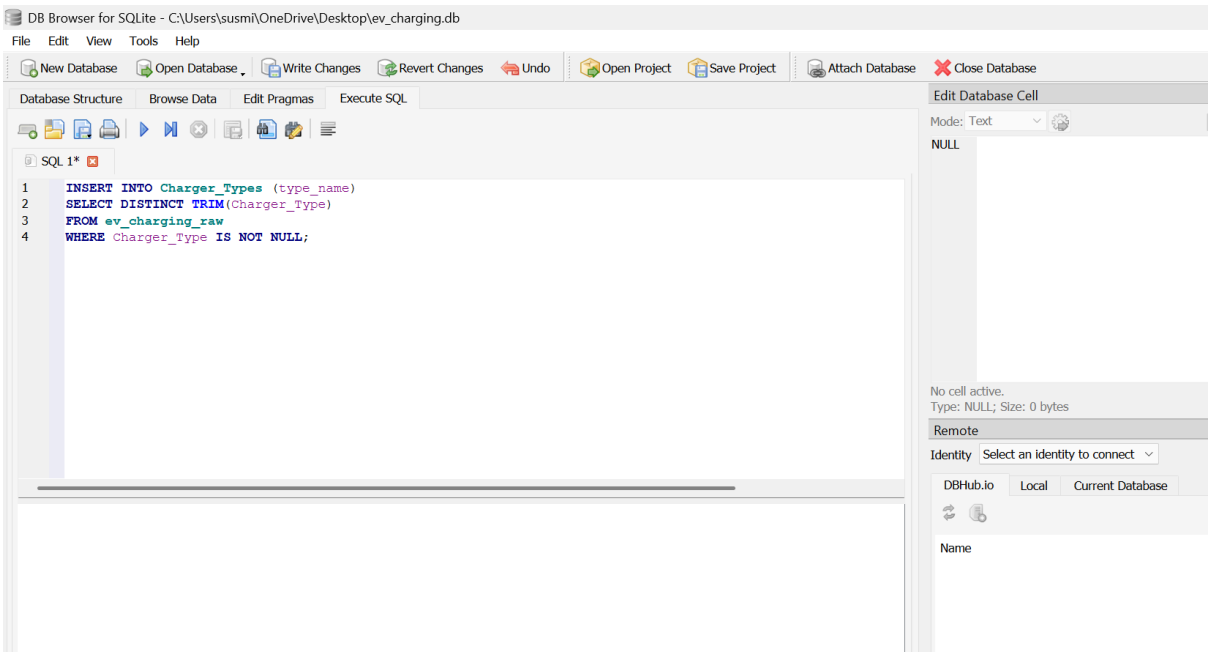
CREATE TABLE Payments (
    payment_id INTEGER PRIMARY KEY AUTOINCREMENT,
    session_id TEXT UNIQUE NOT NULL,
    payment_method TEXT,
    amount_inr REAL,
    FOREIGN KEY (session_id)
        REFERENCES Charging_Sessions(session_id)
);
```



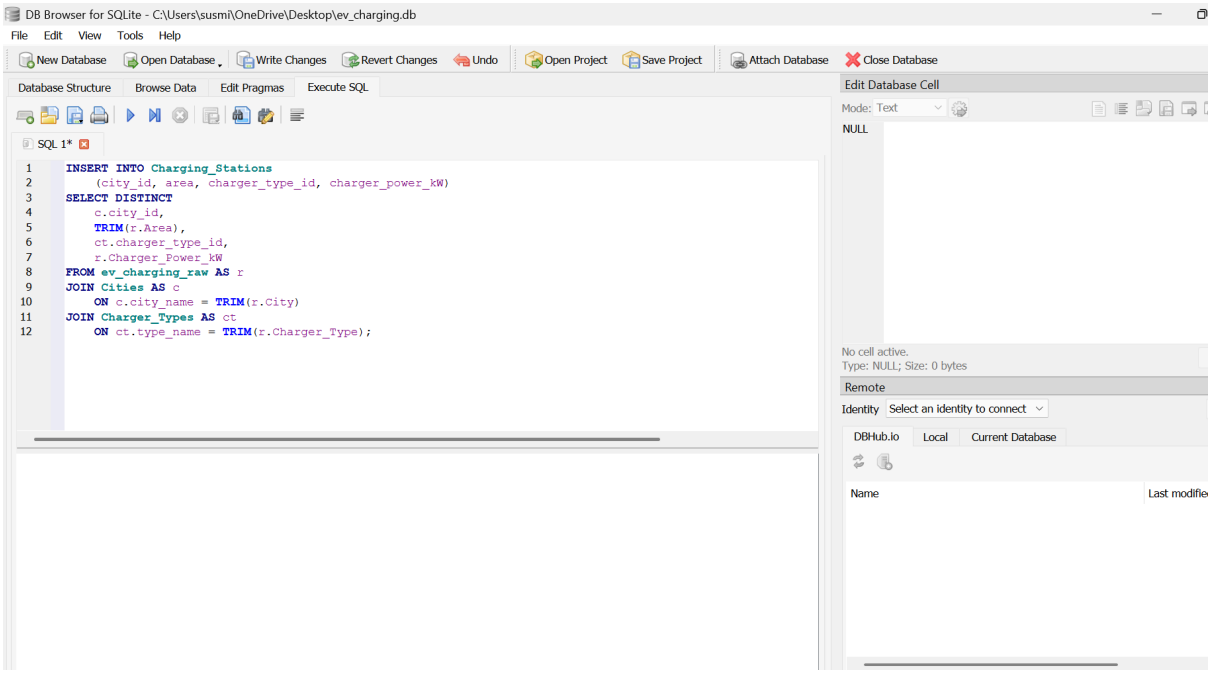

```sql
INSERT INTO Cities (city_name)
SELECT DISTINCT TRIM(City)
FROM ev_charging_raw
WHERE City IS NOT NULL;
```



```sql
INSERT INTO Charger_Types (type_name)
SELECT DISTINCT TRIM(Charger_Type)
FROM ev_charging_raw
WHERE Charger_Type IS NOT NULL;
```

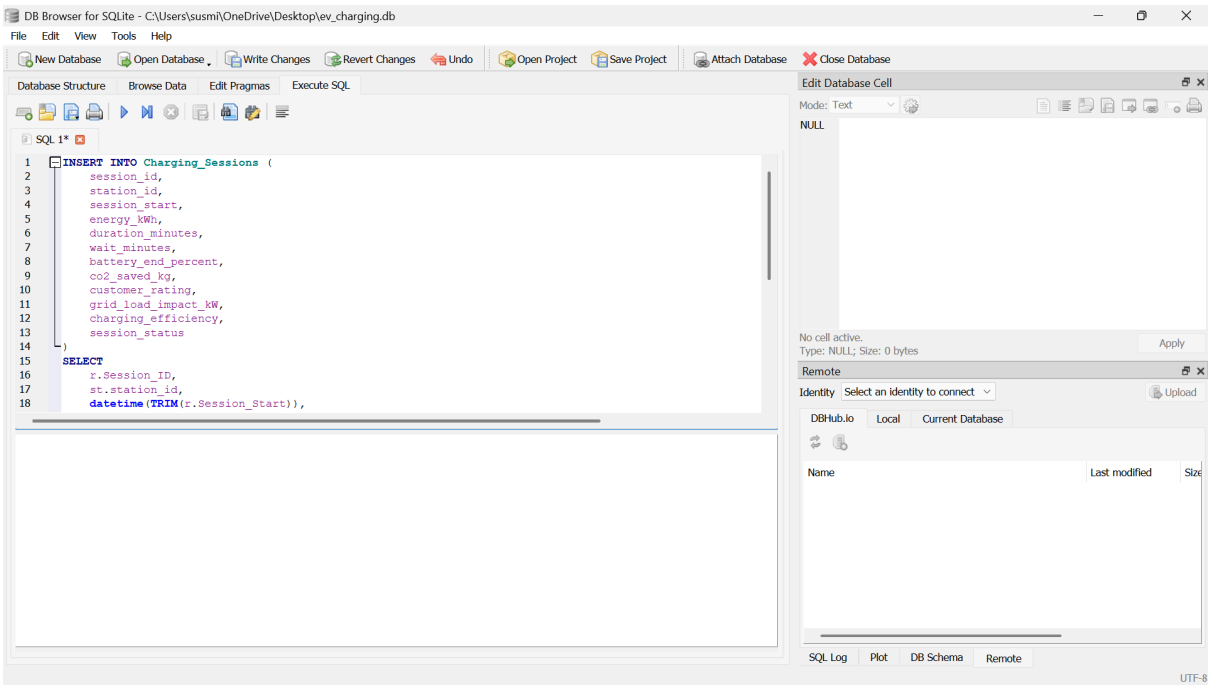



```sql
INSERT INTO Charging_Stations
    (city_id, area, charger_type_id, charger_power_kW)
SELECT DISTINCT
    c.city_id,
    TRIM(r.Area),
    ct.charger_type_id,
    r.Charger_Power_kW
FROM ev_charging_raw AS r
JOIN Cities AS c
    ON c.city_name = TRIM(r.City)
JOIN Charger_Types AS ct
    ON ct.type_name = TRIM(r.Charger_Type);
```



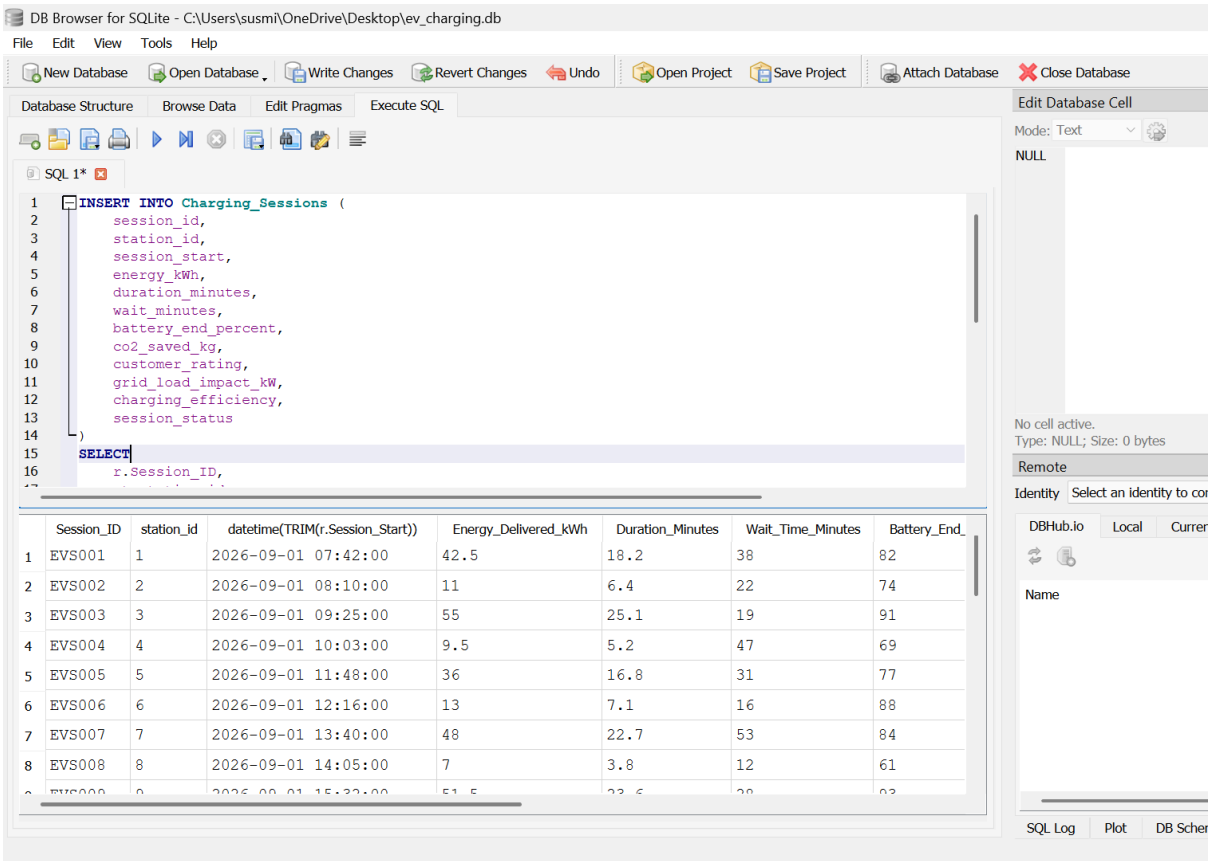

```sql
INSERT INTO Charging_Sessions (
    session_id,
    station_id,
    session_start,
    energy_kWh,
    duration_minutes,
    wait_minutes,
    battery_end_percent,
    co2_saved_kg,
    customer_rating,
    grid_load_impact_kW,
    charging_efficiency,
    session_status
)
SELECT
    r.Session_ID,
    st.station_id,
    datetime(TRIM(r.Session_Start)),
    r.Energy_Delivered_kWh,
    r.Duration_Minutes,
    r.Wait_Time_Minutes,
    r.Battery_End_Percent,
    r.Estimated_CO2_Saved_kg,
    r.Customer_Rating_5,
    r.Grid_Load_Impact_kW,
    r.Charging_Efficiency,
    r.Session_Status
FROM ev_charging_raw AS r
JOIN Cities AS c
```

```
    ON c.city_name = TRIM(r.City)
JOIN Charger_Types AS ct
    ON ct.type_name = TRIM(r.Charger_Type)
JOIN Charging_Stations AS st
    ON st.city_id = c.city_id
    AND st.area = TRIM(r.Area)
    AND st.charger_type_id = ct.charger_type_id
    AND st.charger_power_kW = r.Charger_Power_kW;
```

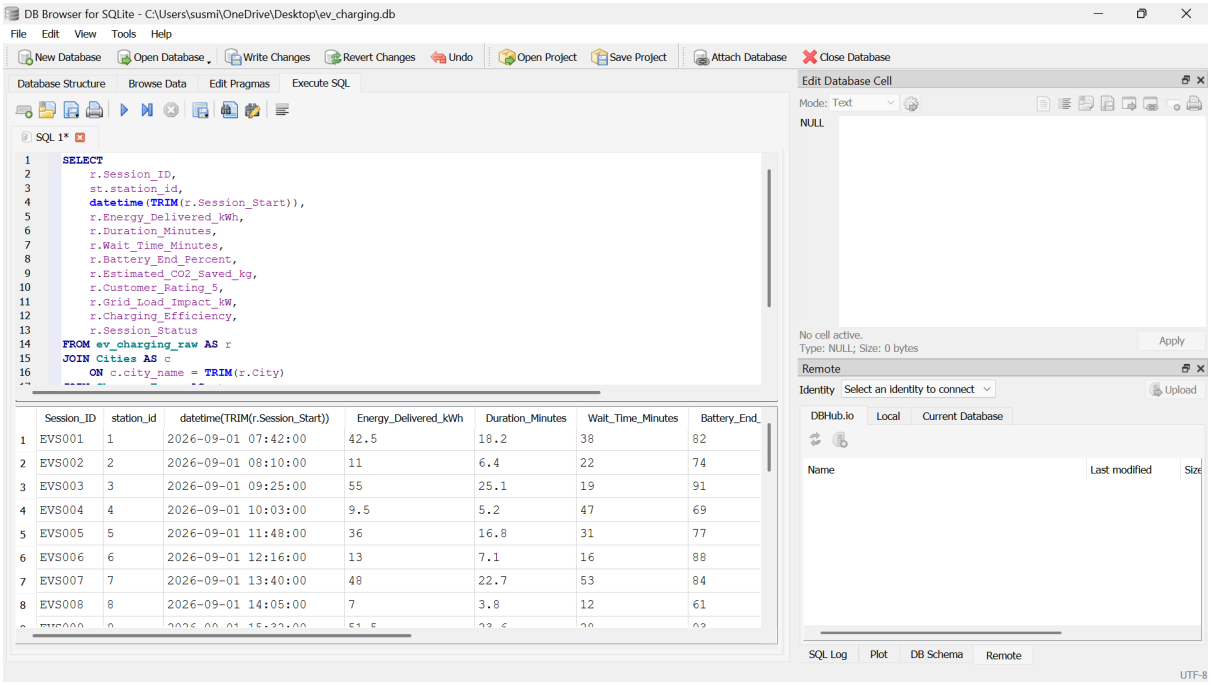



```
SQL
INSERT INTO Charging_Sessions (
    session_id,
    station_id,
    session_start,
    energy_kWh,
    duration_minutes,
    wait_minutes,
    battery_end_percent,
    co2_saved_kg,
    customer_rating,
    grid_load_impact_kW,
    charging_efficiency,
    session_status
)
SELECT
    r.Session_ID,
```

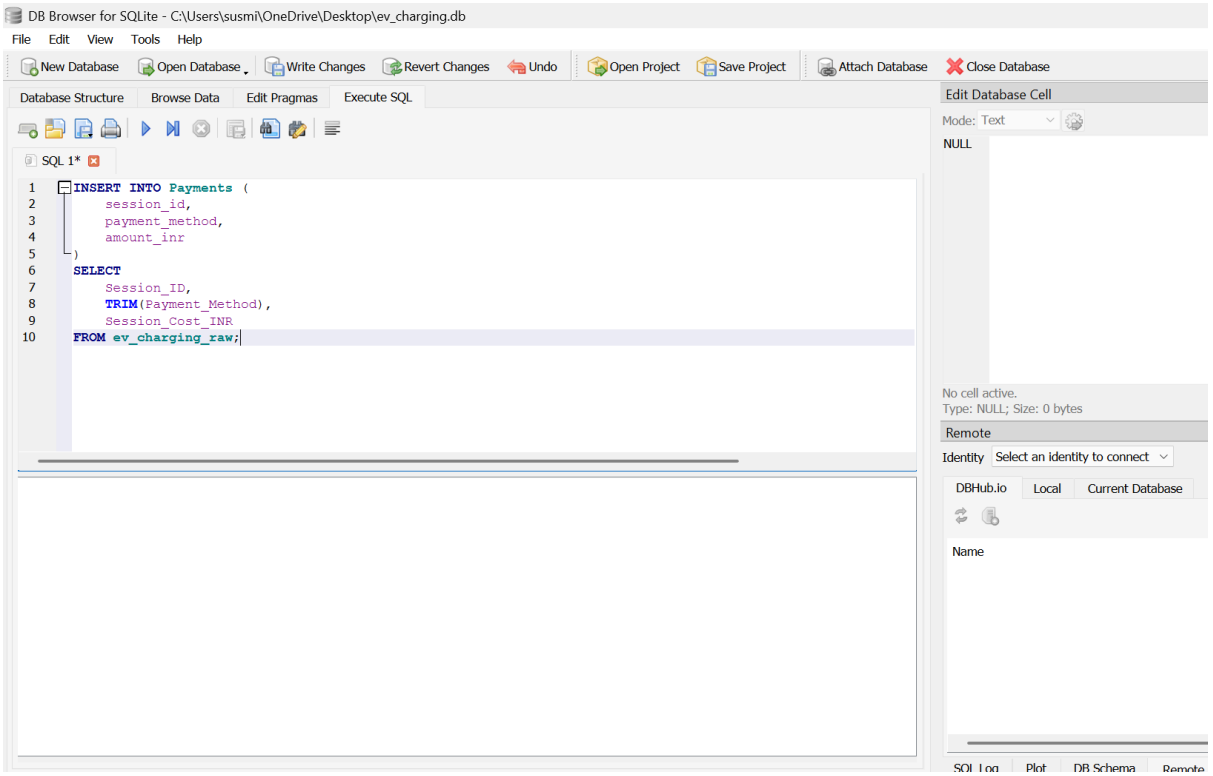
```
    st.station_id,
    datetime(TRIM(r.Session_Start)),
    r.Energy_Delivered_kWh,
    r.Duration_Minutes,
    r.Wait_Time_Minutes,
    r.Battery_End_Percent,
    r.Estimated_CO2_Saved_kg,
    r.Customer_Rating_5,
    r.Grid_Load_Impact_kW,
    r.Charging_Efficiency,
    r.Session_Status
FROM ev_charging_raw AS r
JOIN Cities AS c
    ON c.city_name = TRIM(r.City)
JOIN Charger_Types AS ct
    ON ct.type_name = TRIM(r.Charger_Type)
JOIN Charging_Stations AS st
    ON st.city_id = c.city_id
    AND st.area = TRIM(r.Area)
    AND st.charger_type_id = ct.charger_type_id
    AND st.charger_power_kW = r.Charger_Power_kW;
```



	Session_ID	station_id	datetime(TRIM(r.Session_Start))	Energy_Delivered_kWh	Duration_Minutes	Wait_Time_Minutes	Battery_End_
1	EVS001	1	2026-09-01 07:42:00	42.5	18.2	38	82
2	EVS002	2	2026-09-01 08:10:00	11	6.4	22	74
3	EVS003	3	2026-09-01 09:25:00	55	25.1	19	91
4	EVS004	4	2026-09-01 10:03:00	9.5	5.2	47	69
5	EVS005	5	2026-09-01 11:48:00	36	16.8	31	77
6	EVS006	6	2026-09-01 12:16:00	13	7.1	16	88
7	EVS007	7	2026-09-01 13:40:00	48	22.7	53	84
8	EVS008	8	2026-09-01 14:05:00	7	3.8	12	61




```sql
INSERT INTO Payments (
    session_id,
    payment_method,
    amount_inr
)
SELECT
    Session_ID,
    TRIM(Payment_Method),
    Session_Cost_INR
FROM ev_charging_raw;
```




```sql
SELECT 'Cities' AS table_name, COUNT(*) AS total_rows
FROM Cities

UNION ALL

SELECT 'Charger_Types', COUNT(*)
FROM Charger_Types

UNION ALL

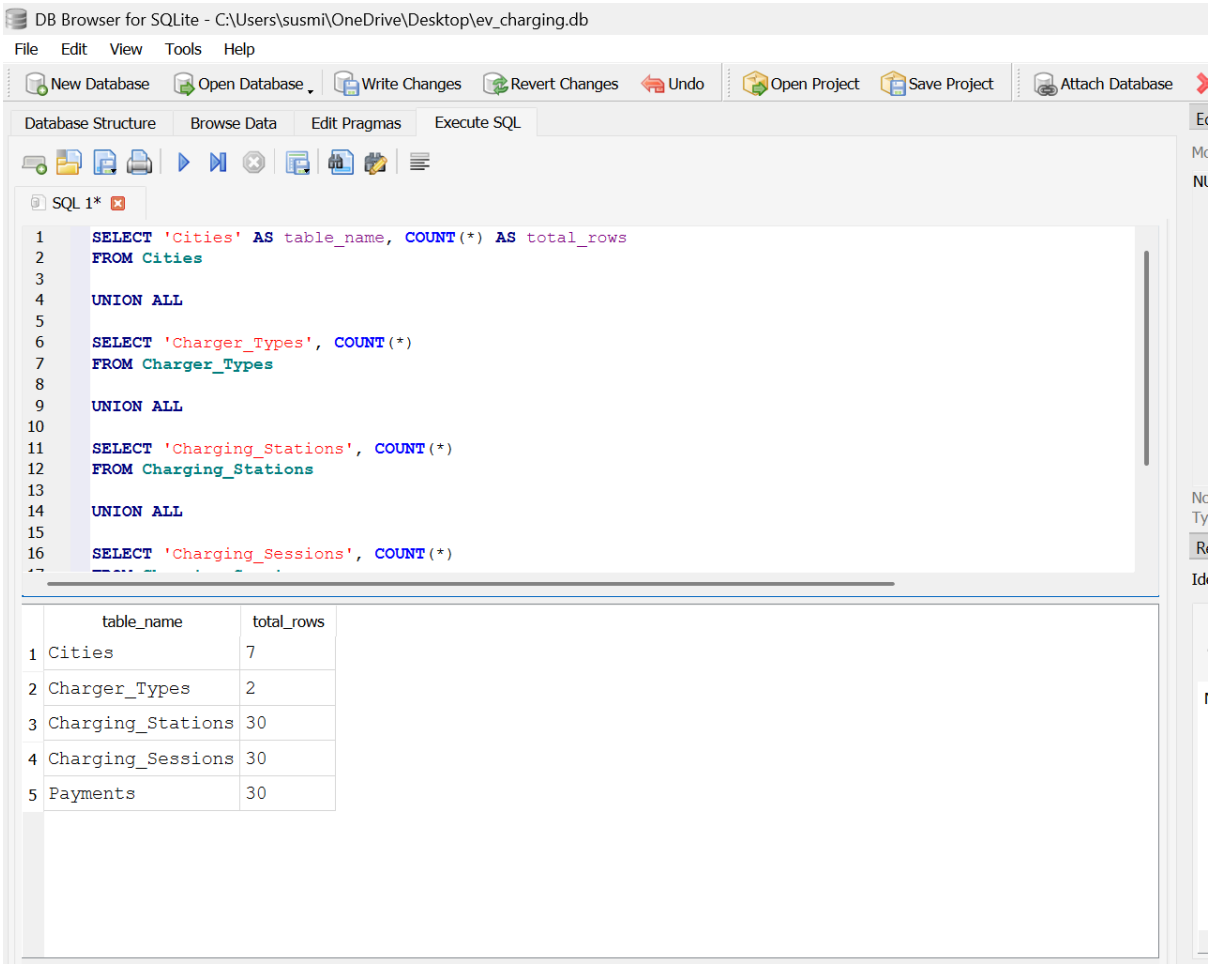
SELECT 'Charging_Stations', COUNT(*)
FROM Charging_Stations

UNION ALL

SELECT 'Charging_Sessions', COUNT(*)
FROM Charging_Sessions

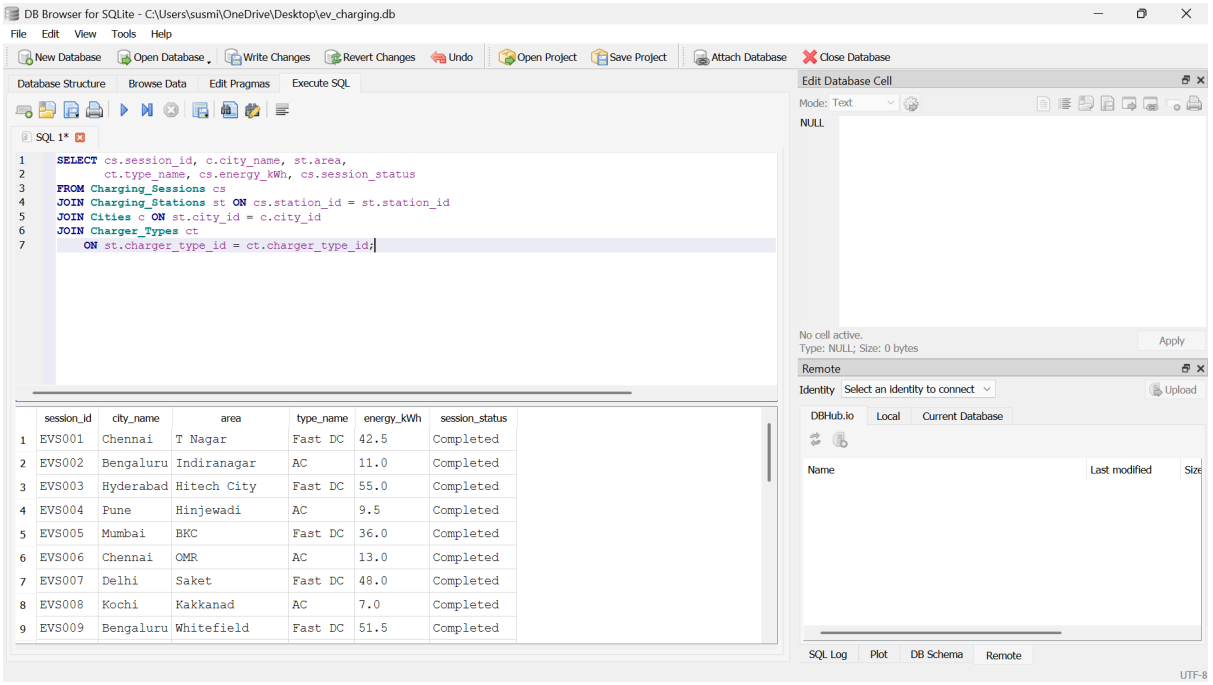
UNION ALL

SELECT 'Payments', COUNT(*)
FROM Payments;
```



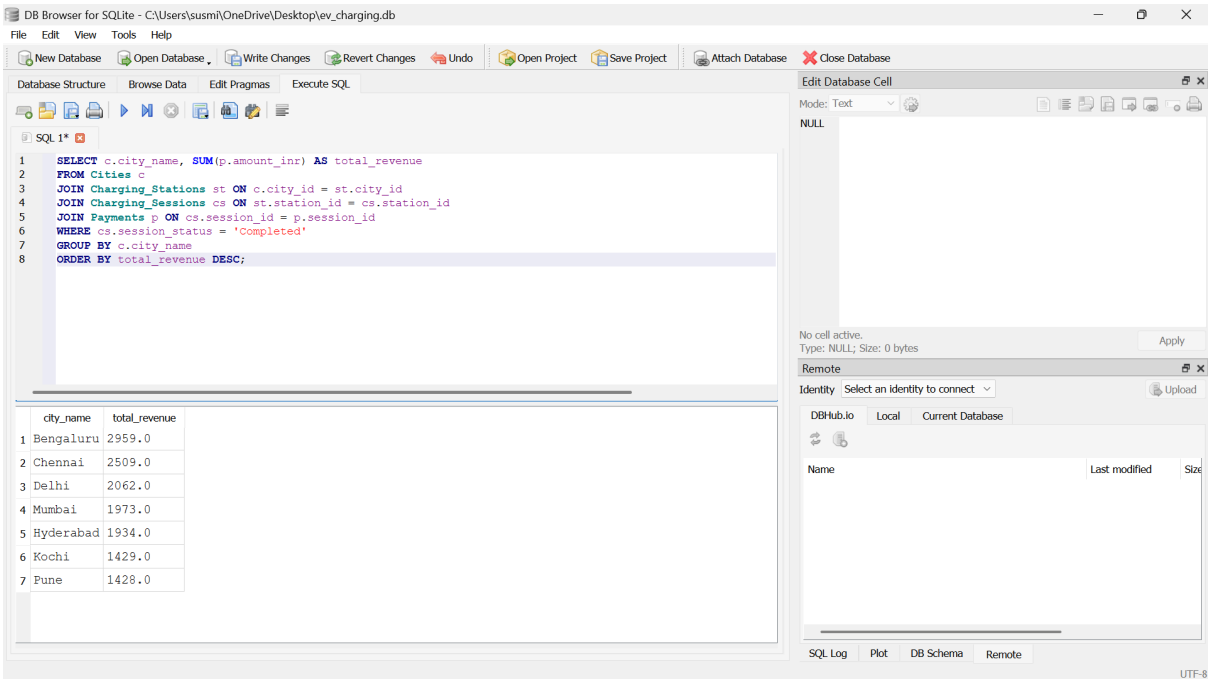
SQL

```sql
SELECT cs.session_id, c.city_name, st.area,
       ct.type_name, cs.energy_kWh, cs.session_status
FROM Charging_Sessions cs
JOIN Charging_Stations st ON cs.station_id = st.station_id
JOIN Cities c ON st.city_id = c.city_id
JOIN Charger_Types ct
    ON st.charger_type_id = ct.charger_type_id;
```

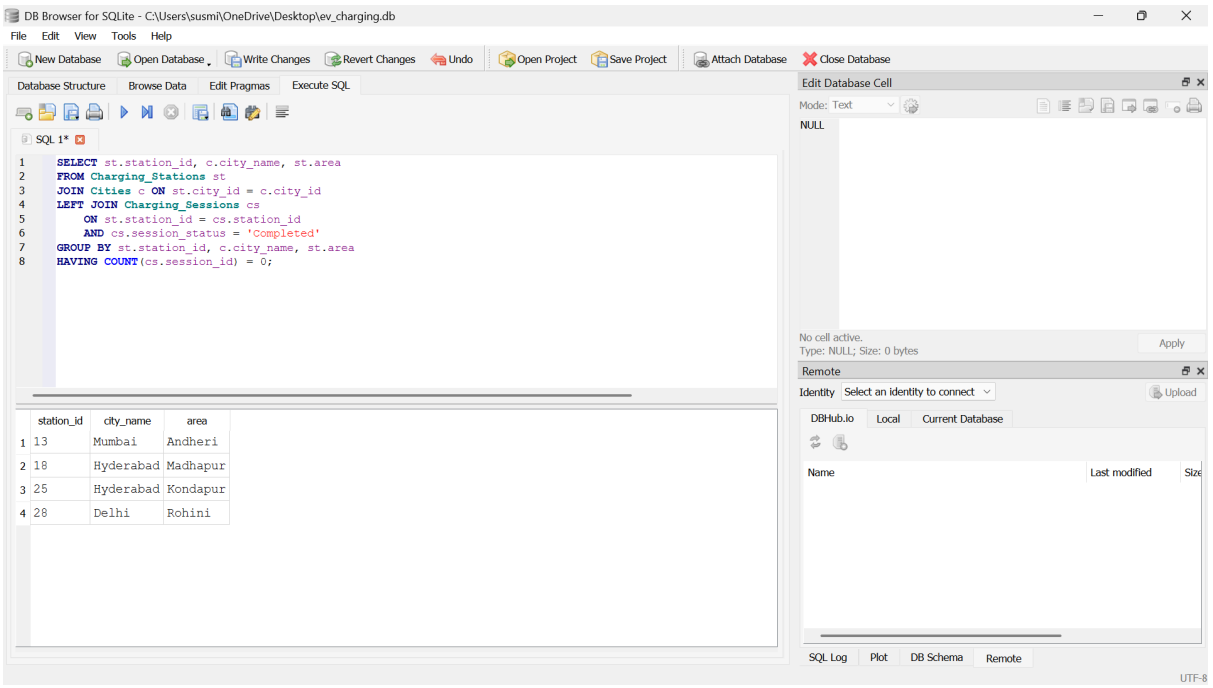


SQL

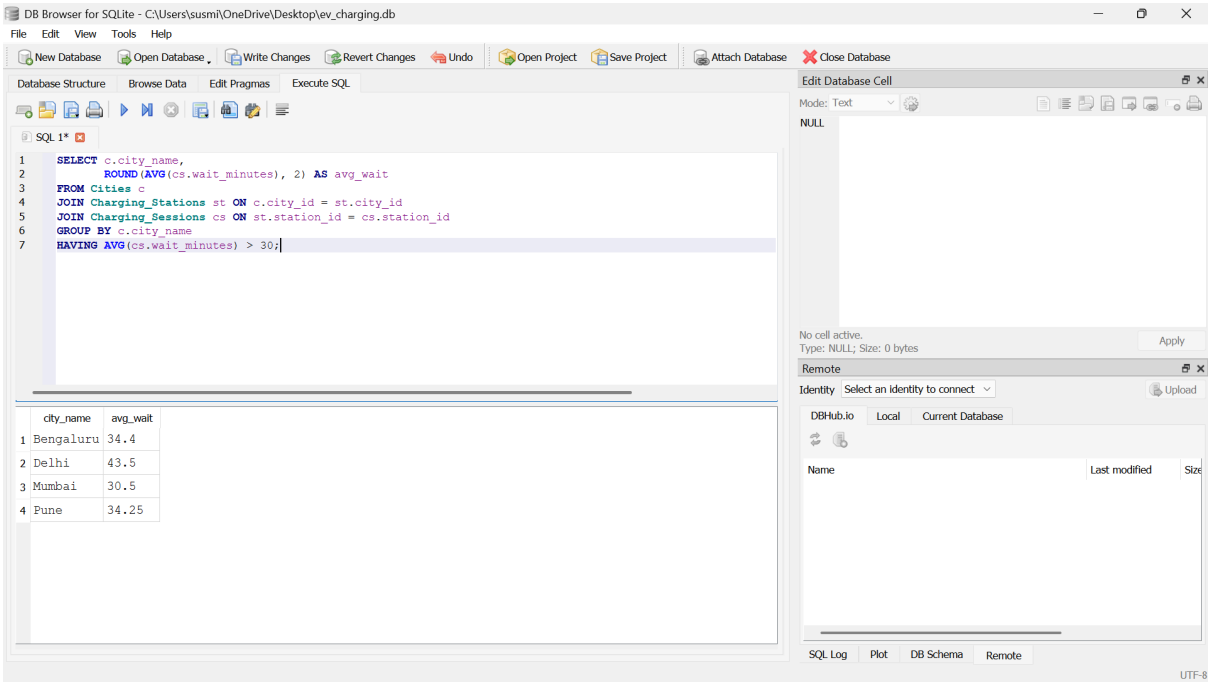
```sql
SELECT c.city_name, SUM(p.amount_inr) AS total_revenue
FROM Cities c
JOIN Charging_Stations st ON c.city_id = st.city_id
JOIN Charging_Sessions cs ON st.station_id = cs.station_id
JOIN Payments p ON cs.session_id = p.session_id
WHERE cs.session_status = 'Completed'
GROUP BY c.city_name
ORDER BY total_revenue DESC;
```



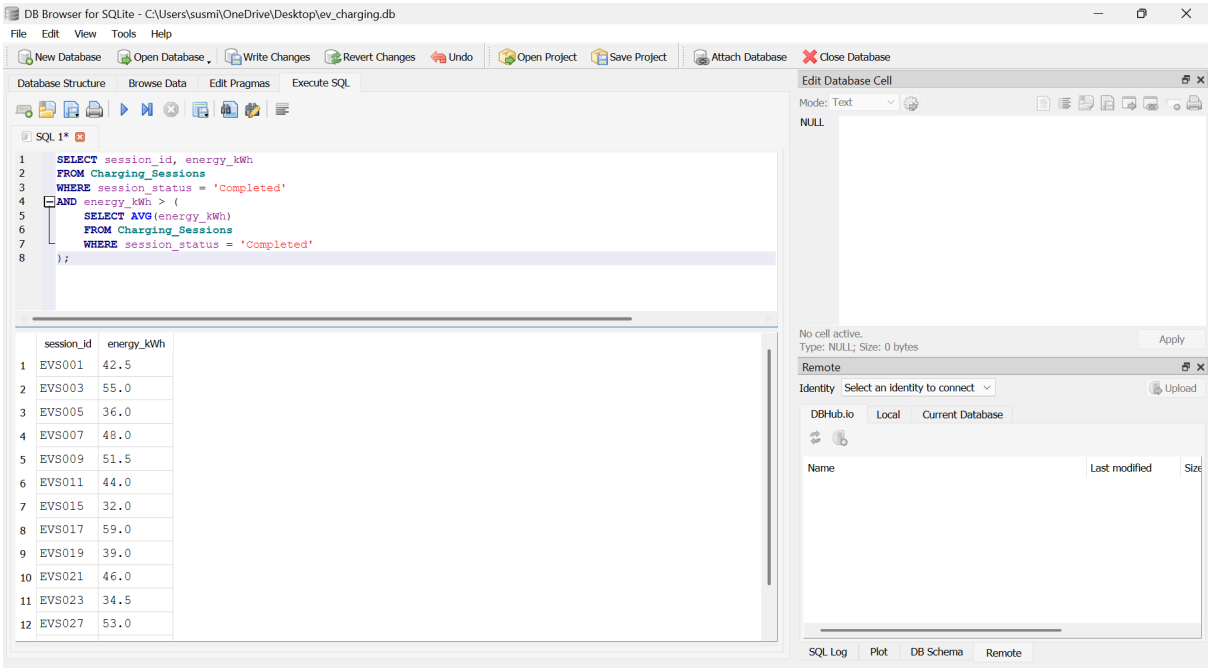
```sql
SELECT st.station_id, c.city_name, st.area
FROM Charging_Stations st
JOIN Cities c ON st.city_id = c.city_id
LEFT JOIN Charging_Sessions cs
    ON st.station_id = cs.station_id
    AND cs.session_status = 'Completed'
GROUP BY st.station_id, c.city_name, st.area
HAVING COUNT(cs.session_id) = 0;
```



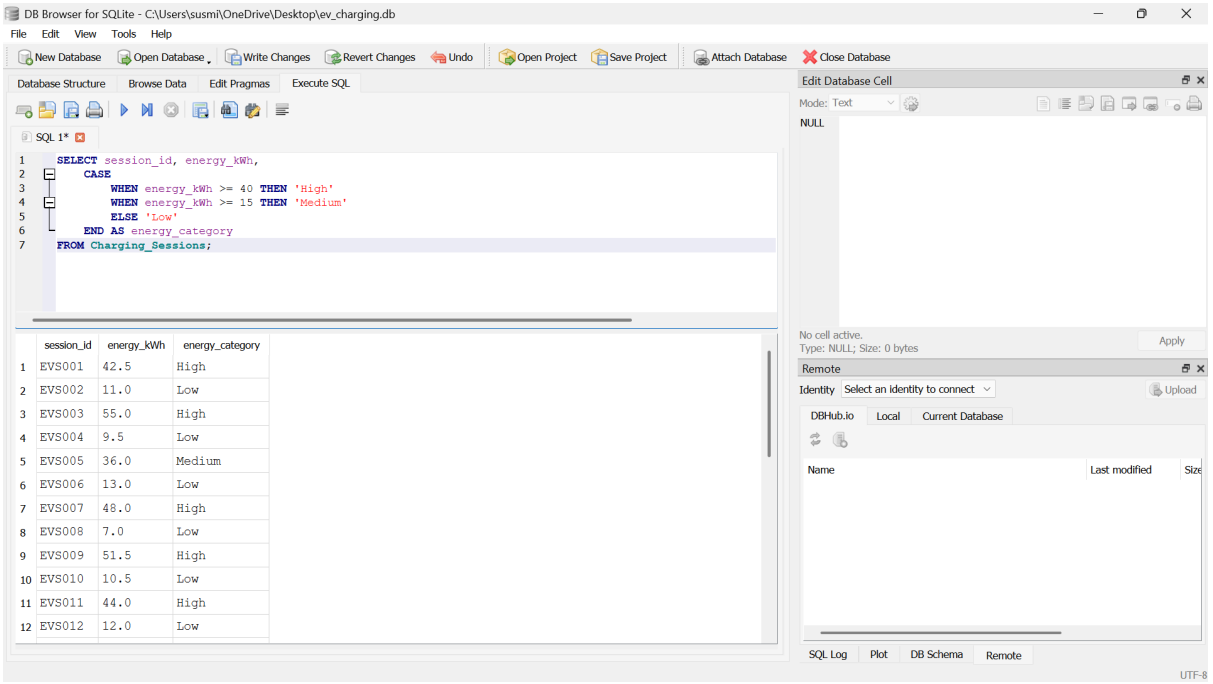

```sql
SELECT c.city_name,
       ROUND(AVG(cs.wait_minutes), 2) AS avg_wait
FROM Cities c
JOIN Charging_Stations st ON c.city_id = st.city_id
JOIN Charging_Sessions cs ON st.station_id = cs.station_id
GROUP BY c.city_name
HAVING AVG(cs.wait_minutes) > 30;
```



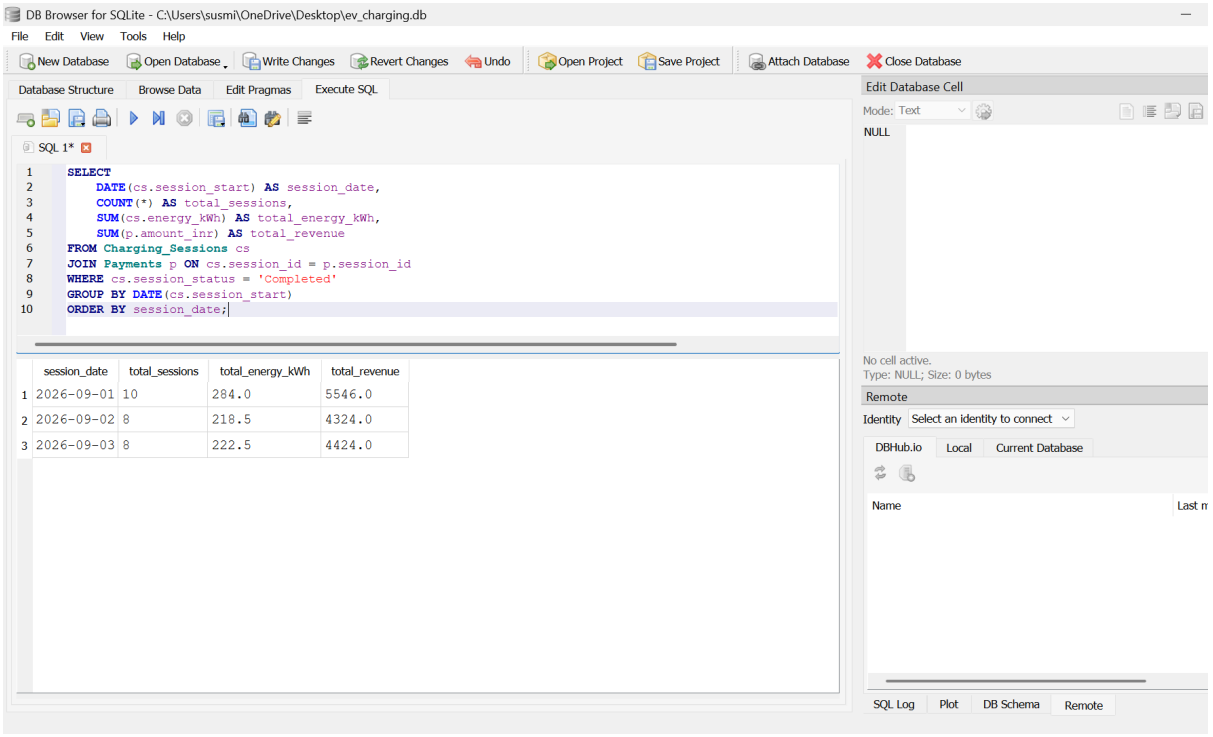

```sql
SELECT session_id, energy_kWh
FROM Charging_Sessions
WHERE session_status = 'Completed'
AND energy_kWh > (
    SELECT AVG(energy_kWh)
    FROM Charging_Sessions
    WHERE session_status = 'Completed'
);
```



```sql
SELECT session_id, energy_kWh,
    CASE
        WHEN energy_kWh >= 40 THEN 'High'
        WHEN energy_kWh >= 15 THEN 'Medium'
        ELSE 'Low'
    END AS energy_category
FROM Charging_Sessions;
```




```sql
SELECT
    DATE(cs.session_start) AS session_date,
    COUNT(*) AS total_sessions,
    SUM(cs.energy_kWh) AS total_energy_kWh,
    SUM(p.amount_inr) AS total_revenue
FROM Charging_Sessions cs
JOIN Payments p ON cs.session_id = p.session_id
WHERE cs.session_status = 'Completed'
GROUP BY DATE(cs.session_start)
ORDER BY session_date;
```


DB Browser for SQLite - C:\Users\susmi\OneDrive\Desktop\ev_charging.db
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
SQL 1*
1 SELECT
2 DATE(cs.session_start) AS session_date,
3 COUNT(*) AS total_sessions,
4 SUM(cs.energy_kWh) AS total_energy_kWh,
5 SUM(p.amount_inr) AS total_revenue
6 FROM Charging_Sessions cs
7 JOIN Payments p ON cs.session_id = p.session_id
8 WHERE cs.session_status = 'Completed'
9 GROUP BY DATE(cs.session_start)
10 ORDER BY session_date;
session_date total_sessions total_energy_kWh total_revenue
1 2026-09-01 10 284.0 5546.0
2 2026-09-02 8 218.5 4324.0
3 2026-09-03 8 222.5 4424.0
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Remote
Identity Select an identity to connect
DBHub.io Local Current Database
Name
SQL Log Plot DB Schema Remote